

## 第五周——日期、字符与特殊值处理

### 题目目的

- (一) 掌握日期型数据的处理与操作。
- (二) 掌握正则表达式与字符串检索。
- (三) 掌握特殊值处理方法。

### 题目

#### 题目一：日期和时间操作。创建脚本文件test0501.R，在脚本文件中完成下面操作。

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- (1) 用 Sys.Date() 和 Sys.Time() 函数分别获取当前的日期和时间，分别赋值给dt和tm，然后用 print函数显示它们的值，用class显示它们的对象类型。验证dt和tm分别减去30的结果，并用注释语句解释结果的意义。
- (2) 用date()函数获取当前系统的日期赋值给dts，用适当的函数显示其对象类型，验证它减去30的结果，解释出现错误的原因。
- (3) 定义向量x为c('2025-3-31','2025-3-24')，用as.Date把x转换成日期型数据，并赋值给x.date，计算两个日期相差的天数。
- (4) 定义向量y为c('2025-3-24;9:20:45','2025-3-24;9:10:45')，试用as.Date将y转换日期型数据，然后计算它们的差；试用strptime转换为时间型数据，然后计算它的差，并用注释语句解释as.Date和strptime两个函数的有什么不同。
- (5) dt1 = "2025-3-24 8:30:15", dt2 = "2025-3-24 10:30:15",直接用difftime计算两者相差多少小时。
- (6) 用strptime转换dt1和dt2后，再用difftime计算它们相差多少小时；并验证使用as.Date函数转换后，再用difftime计算它们相差多少小时，可否能得到正确的结果？

#### 题目二：paste和paste0函数、转义字符。新脚本文件test0502.R，完成下面任务。

---

- (1) 请选择适当函数创建如下图所示的字符串向量x：

```
[1] "a-1" "b-2" "c-3" "d-4" "e-5" "f-6" "g-7" "h-8" "i-9" "j-10"
[11] "k-11" "l-12" "m-13" "n-14" "o-15" "p-16" "q-17" "r-18" "s-19" "t-20"
[21] "u-21" "v-22" "w-23" "x-24" "y-25" "z-26" "a-27" "b-28" "c-29" "d-30"
[31] "e-31" "f-32" "g-33" "h-34" "i-35" "j-36" "k-37" "l-38" "m-39" "n-40"
[41] "o-41" "p-42" "q-43" "r-44" "s-45" "t-46" "u-47" "v-48" "w-49" "x-50"
[51] "y-51" "z-52"
```

(2) 请选择适当函数创建如下图所示的字符串向量y:

```
[1] "a-1" "a-2" "b-3" "b-4" "c-5" "c-6" "d-7" "d-8" "e-9" "e-10"
[11] "f-11" "f-12" "g-13" "g-14" "h-15" "h-16" "i-17" "i-18" "j-19" "j-20"
[21] "k-21" "k-22" "l-23" "l-24" "m-25" "m-26" "n-27" "n-28" "o-29" "o-30"
[31] "p-31" "p-32" "q-33" "q-34" "r-35" "r-36" "s-37" "s-38" "t-39" "t-40"
[41] "u-41" "u-42" "v-43" "v-44" "w-45" "w-46" "x-47" "x-48" "y-49" "y-50"
[51] "z-51" "z-52"
```

(3) 用一次cat函数在命令窗口中输出如下形状的内容:

远上寒山石径斜，白天生处有人家。  
停车坐爱枫林晚，霜叶红于二月花。

## 题目三：使用正则表达式。打开脚本文件test0503.R，完成以下操作。

- (1) 找出含有es或se的单词。
- (2) 找出结尾是es或se的单词。
- (3) 找出以大写英文字母开头的单词。
- (4) 找出开头是小写英文字母，而其他位置含有大写字母的单词。
- (5) 找出含有非英文字母的单词。
- (6) 找出包含有数字的单词。
- (7) 找出有连续重叠字母的单词，不区分大小写。

```
x <- scan('Solomon.txt',
          what = '',
          quote = "",
          fileEncoding = 'gb2312')
```

## 题目四：sub、gsub函数。打开脚本文件test0504.R，完成下面操作。

---

- (1) 使用grep函数把x中的含有单引号、双引号、问号、句号、逗号、分号、冒号和感叹号的单词找出，并显示单词本身。
- (2) 删除字符串中的“与标点符号 (.,?;:!)", 并把结果赋值给y。

```
x<-scan('Solomon2.txt',  
        what = '',  
        quote = "",  
        fileEncoding = 'gb2312')
```

## 题目五：grepl、regexpr、gregexpr函数。打开脚本文件test0505.R，完成下面操作。

---

- (1) 用grepl找出向量x中包含an字符且出现2次及2次以上的字符串
- (2) 用regexpr找出向量x中包含an字符的字符串。
- (3) 用gregexpr找出向量x中包含an字符的字符串
- (4) 请注释语句回答grep、grepl、regexpr、gregexpr这四个函数的返回结果有什么不同？

```
x = c('these are bananas and oranges',  
      'these are apples and ...',  
      'these are peaches')
```

## 题目六：substr函数和substring函数。创建脚本文件test0506.R，完成下面操作。

---

- (1) 用substr函数和substring函数“R is a programming language for statistical computing and graphics”中第8个字符开始长度为7的子字符串。
- (2) 把字符串“R is a programming language for statistical computing and graphics”中第8至第18个字符修改为“程序设计”，观察修改结果，可得出什么结论？

## 题目七：strsplit函数。打开脚本文件test0507.R，完成下面操作。

---

- (1) 把向量x合并为一个字符串，赋值给y。
- (2) 对字符串y进行分词操作，把操作结果赋值给z。

- (3) 删除z中的空字符串。
- (4) 统计w中各个单词出现的频数。

```
x = readLines("Solomon2.txt")
```

## 题目八：特殊数据处理。创建脚本文件test0508.R，完成下面操作。

---

- (1) 创建向量x，其元素为NaN、1、NA、3、Inf、5、NULL，用关系表达式判断向量x的长度是否等于7。
- (2) 分别对NaN、NA、Inf和NULL进行处理，实现以下目标：
- 删除向量中的NaN和NA值。
  - 删除向量中Inf值。

# 答案及解析

## 题目一：

---

```
dt <- Sys.Date()
tm <- Sys.time()
print(dt)
```

[1] "2025-08-30"

```
print(tm)
```

[1] "2025-08-30 19:44:44 CST"

```
class(dt)
```

[1] "Date"

```
class(tm)
```

[1] "POSIXct" "POSIXt"

```
dt-30 #日期减30
```

[1] "2025-07-31"

```
tm-30 #秒数减30
```

[1] "2025-08-30 19:44:14 CST"

```
dts <- date()
class(dts)
```

[1] "character"

```
#dts - 30
#错误于 dts - 30: 二进列运算符中有非数值参数

x <- c('2025-3-31','2025-3-24')
x.date <- as.Date(x)
diff(x.date)
```

Time difference of -7 days

```
y <- c('2025-3-24;9:20:45','2025-3-24;9:10:45')
as.Date(y)
```

```
[1] "2025-03-24" "2025-03-24"
```

```
y.time <- strptime(y,"%Y-%m-%d;%H:%M:%S")
diff(y.time)
```

Time difference of -10 mins

```
dt1 = "2025-3-24 8:30:15"
dt2 = "2025-3-24 10:30:15"
difftime(strptime(dt2, format = "%Y-%m-%d %H:%M:%S"),
         strptime(dt1, format = "%Y-%m-%d %H:%M:%S"), units = "hours")
```

Time difference of 2 hours

```
dt1.time <- strptime(dt1, format = "%Y-%m-%d %H:%M:%S")
dt2.time <- strptime(dt2, format = "%Y-%m-%d %H:%M:%S")
difftime(dt2.time, dt1.time, units = "hours")
```

Time difference of 2 hours

```
dt1.date <- as.Date(dt1)
dt2.date <- as.Date(dt2)
difftime(dt2.date, dt1.date, units = "hours")
```

Time difference of 0 hours

## 题目二：

```
x <- letters
y <- 1:52
paste(x,y,sep = '-')
```

```
[1] "a-1" "b-2" "c-3" "d-4" "e-5" "f-6" "g-7" "h-8" "i-9" "j-10"
[11] "k-11" "l-12" "m-13" "n-14" "o-15" "p-16" "q-17" "r-18" "s-19" "t-20"
[21] "u-21" "v-22" "w-23" "x-24" "y-25" "z-26" "a-27" "b-28" "c-29" "d-30"
[31] "e-31" "f-32" "g-33" "h-34" "i-35" "j-36" "k-37" "l-38" "m-39" "n-40"
[41] "o-41" "p-42" "q-43" "r-44" "s-45" "t-46" "u-47" "v-48" "w-49" "x-50"
[51] "y-51" "z-52"
```

```
a <- rep(letters,each = 2)
paste(a,y,sep = '-')
```

```
[1] "a-1" "a-2" "b-3" "b-4" "c-5" "c-6" "d-7" "d-8" "e-9" "e-10"
[11] "f-11" "f-12" "g-13" "g-14" "h-15" "h-16" "i-17" "i-18" "j-19" "j-20"
[21] "k-21" "k-22" "l-23" "l-24" "m-25" "m-26" "n-27" "n-28" "o-29" "o-30"
[31] "p-31" "p-32" "q-33" "q-34" "r-35" "r-36" "s-37" "s-38" "t-39" "t-40"
[41] "u-41" "u-42" "v-43" "v-44" "w-45" "w-46" "x-47" "x-48" "y-49" "y-50"
[51] "z-51" "z-52"
```

```
x1 <- '远上寒山石径斜，白天生处有人家。\\n停车坐爱枫林晚，霜叶红于二月花。'
cat(x1)
```

远上寒山石径斜，白天生处有人家。  
停车坐爱枫林晚，霜叶红于二月花。

### 题目三:

```
x <- scan('Solomon.txt',
          what = '',
          quote = "",
          fileEncoding = 'gb2312')
```

```
x[grepl('es$',x)]
```

```
[1] "babies" "riches" "enemies" "riches" "slaves"
```

```
x[grepl('se$',x)]
```

```
[1] "house" "Please" "chose" "because" "because" "those" "those"
```

```
x[grepl('^[A-Z]',x)]
```

```
[1] "Long"      "Solomon"   "He"        "In"        "They"      "One"
[7] "The"       "The"       "No"        "The"       "Each"      "So"
[13] "King"      "Solomon"   "Bring"     "King"      "Oh"        "Your"
[19] "Majesty"   "Give"      "Please"     "Then"      "King"      "Solomon"
[25] "Give"      "She"       "God"       "Solomon"   "This"      "I"
[31] "I"         "Why"       "Now"       "I"         "God"       "When"
[37] "Pharaoh"   "Jerusalem" "Solomon"   "I"         "King"      "SoloMon"
[43] "I"         "Solomon"   "Pharaoh's" "I"         "I"         "King"
[49] "Solomon"   "God"       "I"         "I"         "At"        "I"
[55] "They"     "It"        "I"
```

```
x[grepl('^[a-z].*[A-X]',x)]
```

```
[1] "mother"      "youngest"      "conversation"
```

```
x[grepl('^[^a-zA-Z]',x)]
```

```
[1] "women3"      "baby's"      "woman's"      "don't"      "couldn't"      "hadn't"
[7] "so-called"   "Pharaoh's"   "above2"       "don't"
```

```
x[grepl('\\d',x)]
```

```
[1] "women3" "above2"
```

```
x[grepl('([a-z])\\1',x,ignore.case = T)]
```

```
[1] "took"      "quarrel"    "see"        "kill"       "needs"      "good"
[7] "called"    "finally"    "alliance"   "so-called"  "marrying"   "wedding"
[13] "too"       "soon"       "making"     "pulling"    "looks"      "feel"
```

## 题目四:

```
x <- scan('Solomon.txt',
          what = '',
          quote = "",
          fileEncoding = 'gb2312')
```

```
x[grepl('es$',x)]
```

```
[1] "babies" "riches" "enemies" "riches" "slaves"
```

```
x[grepl('se$',x)]
```

```
[1] "house" "Please" "chose" "because" "because" "those" "those"
```

```
x[grepl('^[A-Z]',x)]
```

```
[1] "Long"      "Solomon"   "He"        "In"        "They"      "One"
[7] "The"       "The"       "No"        "The"       "Each"      "So"
[13] "King"      "Solomon"   "Bring"     "King"      "Oh"        "Your"
[19] "MajestTy"  "Give"      "Please"     "Then"      "King"      "Solomon"
[25] "Give"      "She"       "God"       "Solomon"   "This"      "I"
[31] "I"         "Why"       "Now"       "I"         "God"       "When"
[37] "Pharaoh"   "Jerusalem" "Solomon"   "I"         "King"      "SoloMon"
[43] "I"         "Solomon"   "PhaRaoh's" "I"         "I"         "King"
[49] "Solomon"   "God"       "I"         "I"         "At"        "I"
[55] "They"      "It"        "I"
```

```
x[grepl('^[a-z].*[A-X]',x)]
```

```
[1] "moTher"      "younGest"      "converSation"
```

```
x[grepl('^[^a-zA-Z]',x)]
```

```
[1] "women3"      "baby's"      "woman's"      "don't"      "couldn't"      "hadn't"
[7] "so-called"   "PhaRaoh's"   "above2"       "don't"
```

```
x[grepl('\\d',x)]
```

```
[1] "women3" "above2"
```

```
x[grepl('([a-z])\\1',x,ignore.case = T)]
```

```
[1] "took"      "quarrel"     "see"        "kill"       "needs"      "good"
[7] "called"    "finally"     "alliance"   "so-called"  "marrying"   "wedding"
[13] "too"       "soon"        "makiing"    "pulling"    "looks"      "feel"
```

## 题目五:

---

```
x = c('these are bananas and oranges',  
      'these are apples and ...',  
      'these are peaches')  
x[grepl("(an).*\1", x)]
```

```
[1] "these are bananas and oranges"
```

```
regexpr("an", x)
```

```
[1] 12 18 -1  
attr(,"match.length")  
[1] 2 2 -1  
attr(,"index.type")  
[1] "chars"  
attr(,"useBytes")  
[1] TRUE
```

```
gregexpr("an", x)
```

```
[[1]]  
[1] 12 14 19 25  
attr(,"match.length")  
[1] 2 2 2 2  
attr(,"index.type")  
[1] "chars"  
attr(,"useBytes")  
[1] TRUE
```

```
[[2]]  
[1] 18  
attr(,"match.length")  
[1] 2  
attr(,"index.type")  
[1] "chars"  
attr(,"useBytes")  
[1] TRUE
```

```
[[3]]  
[1] -1  
attr(,"match.length")  
[1] -1  
attr(,"index.type")  
[1] "chars"  
attr(,"useBytes")  
[1] TRUE
```

```
# - grep: 返回匹配的元素索引。
# - grepl: 返回逻辑向量，表示每个元素是否匹配。
# - regexpr: 返回第一个匹配项的位置和长度，结果是一个数值向量。
# - gregexpr: 返回所有匹配项的位置和长度，结果是一个列表，每个元素对应一个字符串的所有匹配项。
```

## 题目六:

```
original_string <- "R is a programming language for statistical computing and graphics"
substr_result <- substr(original_string, start = 8, stop = 8 + 7 - 1)
print(substr_result)
```

```
[1] "program"
```

```
substring_result <- substring(original_string, first = 8, last = 8 + 7 - 1)
print(substring_result)
```

```
[1] "program"
```

```
part1 <- substr(original_string, 1, 7)
part2 <- substr(original_string, 19, nchar(original_string))
modified_string <- paste0(part1, "程序设计", part2)
print(modified_string)
```

```
[1] "R is a 程序设计 language for statistical computing and graphics"
```

## 题目七:

```
x = readLines("Solomon2.txt")

y <- as.character(x)

z <- strsplit(y, split = ' ')[[1]]
z[z == ''] <- NA
word_freq <- table(z)
```

## 题目八:

---

```
x = readLines("Solomon2.txt")

y <- as.character(x)

z <- strsplit(y,split = ' ')[[1]]
z[z == ''] <- NA
word_freq <- table(z)
```